Polarimetric SAR Image Classification Based on Hierarchical Scattering-Spatial Interaction Transformer

Jie Geng, *Member, IEEE*, Yuhang Zhang, and Wen Jiang

***Abstract*— How to fully utilize the rich but complex scattering characteristics in polarimetric synthetic aperture radar (PolSAR) data is still a challenge. In this article, a hierarchical scattering-spatial interaction transformer (HSSIT) for polarimetric synthetic aperture radar (SAR) image classification is proposed to effectively combine scattering and spatial characteristics of PolSAR data. The proposed HSSIT adopts a multistage hierarchical structure to extract discriminative features. Specifically, the spatial feature extraction branch (SFEB) is designed to improve the global information perception ability for spatial features, which combines the advantages of the CNN and Transformer to extract local features and capture context dependencies between pixels. A scatter-aware branch (SAB) based on a Transformer is proposed to model the correlation between polarimetric scattering features. Furthermore, we further propose a cross-attention-based information exchange module, which aggregates the tokens from two branches to enhance the discrimination of features for land cover classification. Sufficient experiments are carried out on three widely used PolSAR datasets to certify the effectiveness and superiority of our proposed method.**

***Index Terms*— Polarimetric synthetic aperture radar (SAR) image classification, scattering characteristics, spatial information, Transformer.**

I. Introduction

POLARIMETRIC synthetic aperture radar (PolSAR) is formed by transmitting and receiving radar echoes of multiple polarization combinations from the scattering medium. It obtains echo data under different polarization combinations by operating the radar in different polarimetric modes [1], [2], [3]. Therefore, PolSAR can comprehensively describe the scattering characteristics of ground objects and accurately achieve the inversion of various scattering parameters of ground objects [4], which has been successfully applied to various tasks such as object detection [3], change detection [5], and land cover classification [6]. The land cover classification of the PolSAR image belongs to a pixel-level classification task, which requires identifying each pixel of the image into a relevant land cover category. It is an important task for the interpretation and understanding of PolSAR images.

For PolSAR image classification, useful information to distinguish different land cover categories is of great significance. To obtain a better understanding of polarimetric information, many polarimetric target decomposition algorithms have been proposed, which generally decompose the scattering matrix, coherence matrix, or covariance matrix into different scattering mechanisms. For example, van Zyl [7] decomposed the pixels into various scattering mechanisms, including odd scattering, even scattering, volume scattering, and diffuse scattering. Besides, many other decomposition methods [8], [9], [10] sequentially have been proposed for the understanding of PolSAR data. These target decomposition models can extract the physical characteristics of PolSAR data and possess superior interpretability for land cover classification. However, due to the complicacy of the PolSAR imaging mechanism, such a method that only relies on a target decomposition model makes it difficult to achieve satisfactory classification results on PolSAR datasets.

Machine-learning models have been developed for PolSAR image classification, such as support vector machines (SVMs) [11] and random forest [12]. These models have good generalization ability and can be used as classifiers for PolSAR classification. Li et al. [13] combined the bag of visual words (BoVWs) model with the K-means clustering algorithm to obtain more types of basic features. Moreover, Bayesian theory based on prior knowledge is also applied for synthetic aperture radar (SAR) image classification, such as conditional random fields [14]. However, most of these traditional machine-learning algorithms rely on hand-crafted polarimetric features to classify pixels, and the spatial information of PolSAR image is not fully utilized. It is of great significance to extract effective feature representation from PolSAR data.

In recent years, deep neural networks have been developed rapidly, and deep-learning-based methods have also achieved breakthroughs in remote-sensing images [15]. In particular, CNN-based deep models have been extensively studied and have achieved attractive classification results. A polarimetric-feature-driven 2D-CNN is proposed to extract spatial features and keep good generalization performance for land cover classification [16]. By successively sliding the window, the CNN can extract local features from SAR images. Tan et al. [17] introduced a complex-valued 3D-CNN to improve the extraction capability of spatial features and polarimetric scattering

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The authors are with the School of Electronics and Information, Northwestern Polytechnical University, Xi'an 710129, China (e-mail: gengjie@nwpu.edu.cn).

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features. Shang et al. [18] developed a deep separable convolution to avoid extracting redundant features. Subsequently, various CNN-based feature selection methods have been proposed. For example, Yang et al. [19] designed a 1-D CNN-based module, which can select high-quality features based on Kullback–Leilber divergence between different features. Dong et al. [20] introduced an attention model into the convolutional network, allowing the model to achieve end-to-end feature selection for PolSAR image classification.

Although CNN-based methods still dominate in PolSAR image classification, there are still some unresolved issues. Typically, such patch-based methods take each image patch around the sample pixel and input them into the deep network to predict the category of the center pixel. It is inevitable that there will be other categories of pixels in the patch, which will weaken the class distinguish of spatial features. Moreover, the locality of convolution is not always beneficial. As we all know, convolution performs well in extracting local features, but local connections and weight sharing are the main inductive biases of CNN [21]. It may be the reason for the bottleneck of current CNN-based methods.

Fortunately, the Transformer provides a novel solution for spatial feature learning. This architecture employs a self-attention mechanism to model correlations of different ranges between input sequence elements [22]. Inspired by the success of the Transformer, some studies have introduced a self-attention mechanism into the model for better classification performance. Hua et al. [23] designed an attention-based network to enrich the multiscale spatial information in a way of spatial sequence. Ren and Zhou [24] utilized a residual attention module to obtain discriminative features with various resolutions. Fan et al. [25] proposed an efficient fusion attention unit to learn the correlation between the channels of features. Later, vision transformer (ViT) [26] replaced the word vectors as the input to the Transformer with image patches and modeled the context relationship of interactive image information through an attention mechanism. Recently, ViT was directly used by Wang et al. [27] to model the relationship of pixels in an image patch and achieved better classification performance than CNN-based methods.

Compared with the CNN and its variants, the Transformer possesses the ability to capture context dependencies among pixels and effectively learn global information from PolSAR patches, which has been confirmed in [27]. The attention mechanism of the Transformer allows the network to pay more attention to the regions of interest, which are related to the category of the central pixel. In other words, even if the patch feature contains characteristics from other categories of pixels, the Transformer can still suppress their influences. However, the computational complexity of self-attention is positively correlated with the square of the token number, which can bring a huge amount of parameters to the Transformer.

Currently, numerous patch-based methods have been dedicated to paying too much attention to spatial correlation, which leads to insufficient utilization of polarimetric features. For PolSAR image classification, different polarimetric scattering features are combined to form semantic characteristics that reflect a difference in land cover categories. It indicates that the correlation between polarimetric scattering features corresponding to each type of ground object should be different. Therefore, modeling the interrelationships between polarimetric scattering features is beneficial for obtaining more discriminative features.

To fully explore deep spatial semantic information and polarimetric characteristics from PolSAR images, a hierarchical scattering-spatial interaction Transformer (HSSIT) is proposed for PolSAR image classification. The proposed HSSIT adopts a multistage hierarchical structure to extract discriminative spatial semantic features and scatter-aware features from image patches. First, our proposed scatter-aware branch (SAB) learns the correlation between polarimetric features. Second, our spatial feature extraction branch (SFEB) is structured by iteratively using the CNN and Transformer, which integrates the feature extraction advantages of these two structures. Third, an information exchange module is embedded into each stage of the hierarchical model to aggregate tokens from two branches. The entire network framework is completely attention-based, which is suitable for the aggregation of polarimetric features and spatial features. The main advantages of our proposed method are given in the following.

1) A novel deep feature extraction network combining the advantages of the CNN and Transformer is proposed. It possesses both the ability to model different range context dependencies among pixels and extract local spatial features from PolSAR image patches.
2) A scatter-aware module based on a Transformer is proposed to capture the correlation between different polarimetric scattering features, allowing the model to adaptively obtain more discriminative scatter-aware features.
3) A scattering-spatial interaction module (SSIM) can establish information exchange between two branches through cross attention mechanism, adaptively complementing and enhancing feature representations.

The remainder of this article is organized as below. The description of the related work is arranged in Section II. In Section III, our proposed method is specifically described, including the design details of each component. Then, the comparative experimental results of the proposed methods and ablation studies are reported and discussed in Section IV. Finally, the conclusion of this study is given in Section V.

II. Related Work

In this section, we introduce the related work of the proposed HSSIT, which is applied in the following method.

A. Polarimetric Decomposition for PolSAR Images

PolSAR images can capture rich physical information such as orientation, size, and surface smoothness of target scatters. For all scattering targets, scattering information can be included in the complex scattering matrix $\boldsymbol{S}$, which is expressed as

$$\boldsymbol{S} = \begin{bmatrix} S_{hh} & S_{hv} \\ S_{vh} & S_{vv} \end{bmatrix} \tag{1}$$

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where S_{ij} represents the scattering coefficient under different polarization combinations [28]. According to the reciprocity theorem $S_{hv} = S_{vh}$ [29], the Pauli scattering vector can be defined as follows:

$$\boldsymbol{k}_{3p} = \frac{1}{\sqrt{2}}[S_{hh} + S_{vv}, S_{hh} - S_{vv}, 2S_{hv}]. \tag{2}$$

To estimate the second-order statistics of the signal, the polarimetric coherence matrix $\boldsymbol{T}$ [30] can be obtained based on $\boldsymbol{k}_{3p}$, which can be formulated as

$$[\boldsymbol{T}]_{3\times 3} = \boldsymbol{k}_{3p} \cdot \boldsymbol{k}_{3p}^{*\top} = \begin{bmatrix} T_{11} & T_{12} & T_{13} \\ T_{21} & T_{22} & T_{23} \\ T_{31} & T_{32} & T_{33} \end{bmatrix} \tag{3}$$

where $*$ denotes the conjugate operation. In addition, the polarimetric coherence matrix T formed by the multilook complex (MLC) PolSAR data can be transformed into a 6-D real value feature (RF) [31], which is suitable as input for deep neural networks. The formula for transformed calculation is written as follows:

$$\mathrm{RF}_1 = 10\log_{10}(T_{11} + T_{22} + T_{33}) \tag{4}$$

$$\mathrm{RF}_2 = T_{22}/(T_{11} + T_{22} + T_{33}) \tag{5}$$

$$\mathrm{RF}_3 = T_{33}/(T_{11} + T_{22} + T_{33}) \tag{6}$$

$$\mathrm{RF}_4 = |T_{12}|/\sqrt{T_{11} \cdot T_{22}} \tag{7}$$

$$\mathrm{RF}_5 = |T_{13}|/\sqrt{T_{11} \cdot T_{33}} \tag{8}$$

$$\mathrm{RF}_6 = |T_{23}|/\sqrt{T_{33} \cdot T_{22}} \tag{9}$$

where T_{ii} ($i = 1, 2,$ or 3) denotes the element of the polarimetric coherence matrix T.

Due to the different surface roughness or uneven propagation media, different objects correspond to different scattering mechanisms. Traditional target scattering models include single scattering, diffuse scattering, even scattering, and volume scattering. For example, the dominant scattering mechanism of smooth surfaces such as farmland and water surfaces is single scattering. The surface of forests and shrubs is best described by the volume scattering model. Generally, polarimetric target decomposition methods can be divided into coherent and noncoherent target decomposition. Among them, coherent decomposition models include Pauli, Krogager decomposition [9], and so on. Noncoherent decomposition models include Yamaguchi et al. [32], Freeman [8], $H/A/\alpha$ [33], and so on. These polarimetric decompositions can be regarded as transforming the coherence matrix to another set of parameters.

B. Visual Transformer for Feature Extraction

ViT has exhibited extraordinary performance in various computer vision tasks [34]. ViTs [26] interpret an image as a sequence of patches, which are embedded in the encoder of the Transformer for information processing. Dong et al. [35] replaced the patch embedding layer with pixel embedding and then applied Transformer to PolSAR image classification. The motivation is to capture long-distance interactions between pixels by the attention mechanism, which can obtain more global spatial relations. Inspired by Swin Transformer [36], Jamali et al. [37] employed local window attention to improve the feature generalization of local regions. The key components of ViT are briefly described as follows [26].

1) Multihead Attention Mechanism: First, a patch embedding layer projects the input sequences X into query matrix $\boldsymbol{Q}$, key matrix $\boldsymbol{K}$, and value matrix $\boldsymbol{V}$. After the attention head intersects the query matrix $\boldsymbol{Q}$ with the corresponding key matrix $\boldsymbol{K}$, it assigns weights to the value matrix $\boldsymbol{V}$ based on the obtained attention matrix $\boldsymbol{A}$, thereby updating the output vector. The formula for this process can be described as follows:

$$Z_i = \mathrm{Attention}(\boldsymbol{Q}_i, \boldsymbol{K}_i, \boldsymbol{V}_i) = \mathrm{Softmax}\left(\frac{\boldsymbol{Q}_i {\boldsymbol{K}_i}^T}{\sqrt{d_k}}\right)\boldsymbol{V} \tag{10}$$

where d_k represents the scaling factor. It can make the variance smaller and the gradient update more stable during training. Through multiple attention heads, the multihead self-attention mechanism can fully perceive information by mapping features to multiple feature subspaces, which is formulated as

$$\mathrm{MultiHead} = \mathrm{Concat}(Z_1, Z_2, \ldots, Z_n) \cdot \boldsymbol{W}_i^O \tag{11}$$

where $\boldsymbol{W}_i^O$ denotes the projected matrix, n is the number of head.

2) Positional Encoding: To preserve the location information of the input sequence, a supplemental position vector is added to the input. In a typical encoding method, the *cosine* function can be written as

$$\mathrm{PE}_{(\mathrm{pos},i)} = \begin{cases} \sin(\mathrm{pos} \cdot w_k), & \text{if } i = 2k \\ \cos(\mathrm{pos} \cdot w_k), & \text{if } i = 2k+1 \end{cases}$$
$$w_k = \frac{1}{10000^{2k/d}}, \quad k = 1, \ldots, d/2 \tag{12}$$

where d stands for the length of the input vector and i denotes the index of each element.

3) Feed-Forward Network: This module is connected behind the multihead attention mechanism and receives updated features for forward propagation, which can be expressed as follows:

$$\mathrm{FFN} = \mathrm{RELU}(W_1 x + b_1)W_2 + b_2 \tag{13}$$

where $RELU(\cdot)$ denotes the activation function.

C. Integration of Polarimetric Information

Currently, more studies have focused on the effect of scattering characteristics of PolSAR data based on deep neural networks [38], [39], [40]. Huang et al. [41] designed an explainable model to generate abstract physical representation from scattering features. Then, this semantic level representation is injected into the neural network in a feature concatenation manner, allowing the model to yield superior performance for PolSAR image classification. Besides, some feature fusion frameworks are proposed to combine features extracted by the CNN with polarimetric features of PolSAR data [42], [43]. Cao et al. [44] applied an attention mechanism to capture the relevance and diversity between features, which can effectively eliminate information differences.

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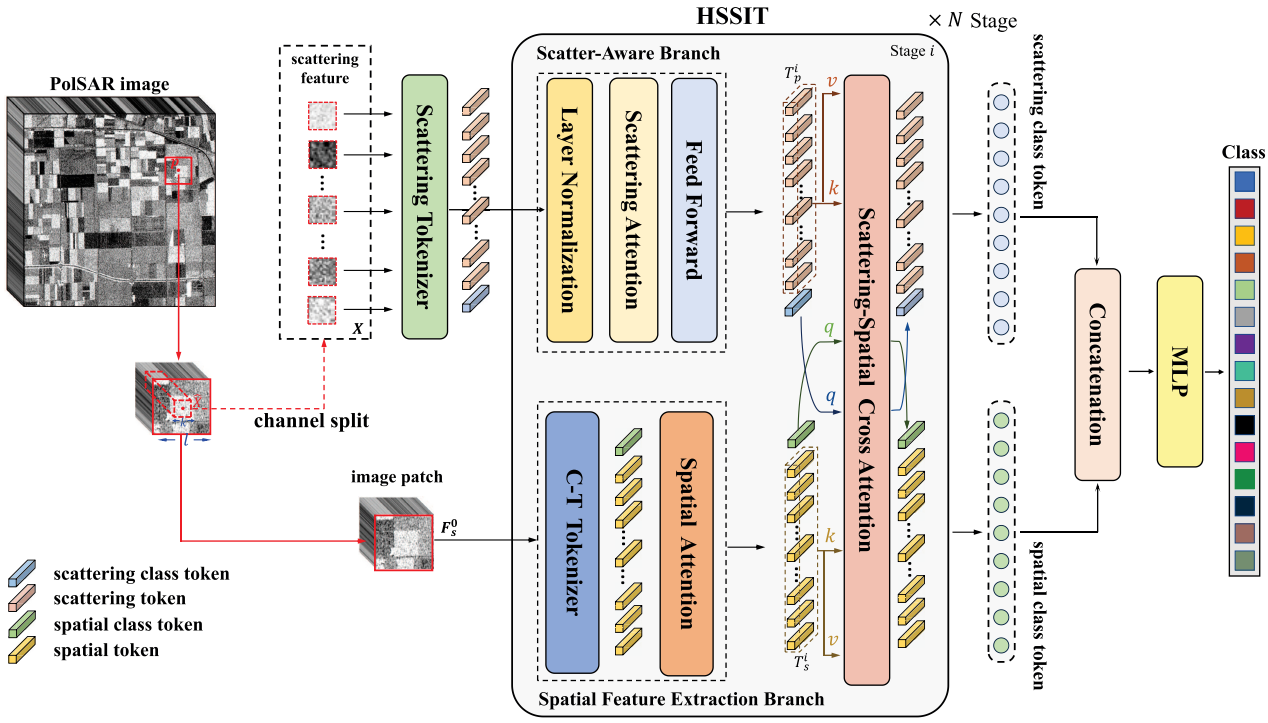


Fig. 1. Overall framework of HSSIT, which consists of N hierarchical stages stacked together. For simplicity and clarity, only one hierarchical stage of HSSIT is shown.

More advanced methods need to be developed to embed scattering characteristics from PolSAR data into the deep neural network for better performance.

III. Method

In our work, an HSSIT is proposed for PolSAR image classification, which is designed to effectively combine polarimetric scattering features and spatial features to generate discriminative deep semantic features. The overall network architecture of HSSIT is shown in Fig. 1, which is constructed by several hierarchical stages. One hierarchical stage is composed of an SAB, an SFEB, and an SSIM. In Sections III-A–III-C, the module we proposed above will be described in detail.

A. Scatter-Aware Branch

For PolSAR data, the input of the model contains rich polarimetric scattering features derived from polarimetric decomposition models, and each channel of input data corresponds to different scattering information. Specifically, Pauli, Krogager, $H/A/\alpha$, and Yamaguchi polarimetric decompositions are applied to obtain rich polarimetric features. These polarimetric features and 6-D scattering features transformed from coherence matrix T are utilized as the input features. Different polarimetric features are combined to form semantic features that reflect category differences. To acquire more discriminative scatter-aware features, an SAB based on the Transformer is developed to model the interrelationships between polarimetric features from image patches, whose structure is shown in Fig. 2. Considering the self-attention mechanism, the Transformer has great potential to handle context dependencies. In this branch, the correlation along the scattering feature dimensions of the PolSAR image patch can be modeled.

In detail, as for a training sample pixel $\boldsymbol{p} \in \mathbb{R}^{1\times1\times C}$ in PolSAR image, we take it as the central pixel and obtain an image patch $\hat{\boldsymbol{X}} \in \mathbb{R}^{k\times k\times C}$ around its neighborhood as the input of SAB. Along the channel direction of this image patch, $\hat{\boldsymbol{X}}$ can be redescribed as $\boldsymbol{X} = \{x_1, x_2, \ldots, x_C\}$, $x_i \in \mathbb{R}^{k\times k}$, where C is the number of scattering characteristic channel, k denotes the size of the image patch. Since the context relationship of polarimetric scattering features is related to the central pixel, k is set to a relatively small size to avoid mixing pixels from other categories into the patch as much as possible. And then, polarimetric feature x_i is projected into a 1-D feature embedding vector $\boldsymbol{t^p}_i \in \mathbb{R}^d$. Here, d stands for the dimension of feature embedding, which is equal to the output channels in the convolution operation. The formula of this process is described as follows:

$$\boldsymbol{t^p}_i = \boldsymbol{W}^p \text{flatten}(x_i), \quad x_i \in \mathbf{X} \tag{14}$$

where $\boldsymbol{W}^p \in \mathbb{R}^{d\times k^2}$ stands for the weight matrix.

The token sequence $\boldsymbol{T_p}$ can be organized by $\boldsymbol{t^p}_i$, that is, $\boldsymbol{T_p} = \{\boldsymbol{t^p}_{\text{cls}}, \boldsymbol{t^p}_1, \boldsymbol{t^p}_2, \ldots, \boldsymbol{t^p}_C\}$. And then, $\boldsymbol{T_p}$ is imported into the Transformer structure, in which $\boldsymbol{t^p}_{\text{cls}}$ represents the class token containing abstract semantic information. In detail, the class token is a single learnable vector that propagates forward along with other tokens mapped from image patches. In the self-attention mechanism, the class token interacts with other tokens to aggregate all features to obtain a global representation [26]. The final output vector of the class token is

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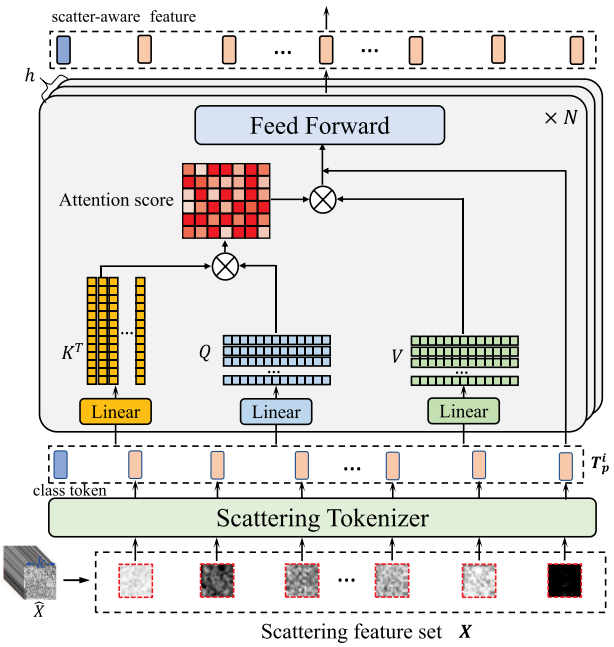


Fig. 2. Schematic of SAB.

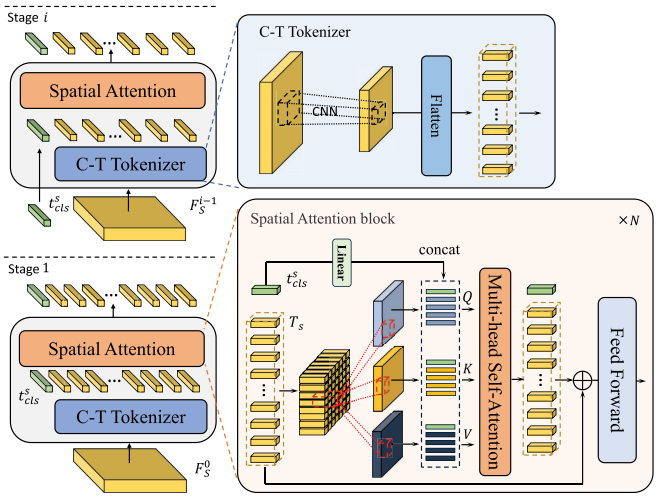


Fig. 3. Structure diagram of SFEB.

used for classification, thereby avoiding prediction bias toward a specific token in the token sequence.

After that, the embedded token sequence $\boldsymbol{T_p}$ is fed into the self-attention mechanism, in which the dependent relationship between the embedded vectors corresponding to each scattering feature can be modeled. The mathematical description of this process can be written as

$$\boldsymbol{q}_i = \boldsymbol{W}^q \boldsymbol{t^p}_i, \quad \boldsymbol{k}_i = \boldsymbol{W}^k \boldsymbol{t^p}_i, \quad \boldsymbol{v}_i = \boldsymbol{W}^v \boldsymbol{t^p}_i \tag{15}$$

$$\boldsymbol{A} = [\boldsymbol{a}_{ij}]_{C\times C} = \left[\frac{\boldsymbol{q}_i{}^T \boldsymbol{k}_i}{\|\boldsymbol{q}_i\|\|\boldsymbol{k}_i\|}\right]_{C\times C} \tag{16}$$

$$\boldsymbol{Y} = \boldsymbol{A}\boldsymbol{V} \tag{17}$$

where $\boldsymbol{a}_{ij}$ denotes the attention score, which describes the correlation between feature vector $\boldsymbol{t^p}_i$ and feature vector $\boldsymbol{t^p}_j$. $\boldsymbol{V}$ is the matrix composed by $\boldsymbol{v}_i$. $\boldsymbol{Y}$ is the output matrix of the attention mechanism.

After contextual modeling of polarimetric scattering features based on the self-attention mechanism, the class token $\boldsymbol{t^p}_{\text{cls}}$ is considered as an advanced semantic adaptive feature that can be inferred from the scattering characteristics of neighborhood pixels. In the subsequent SSIM, this scattering class token $\boldsymbol{t^p}_{\text{cls}}$ will represent the SAB to exchange information with the tokens from the SFEB.

B. Spatial Feature Extraction Branch

Convolution has an advantage in extracting features from local neighborhoods, but it blocks the acquisition of global information, especially in the shallow layers of the network. For land cover classification, the local features and the spatial context relations in the image patch are both of significance. To alleviate these issues, we connect Transformer blocks after the CNN to improve the global modeling of spatial features. Here, an SFEB is designed to combine the advantages of CNN and Transformer, whose structure is depicted in Fig. 3. To obtain more semantic information in the 2-D spatial dimension, we fill more pixels around the input image patch of SAB to get a larger image patch $\boldsymbol{F_s^0}$ with a size of l, which is shown in Fig. 1.

This novel feature extraction branch is composed of a Conv-Transformer tokenizer and spatial attention mechanism. In this branch, tokens are an indispensable composition of the Transformer, which is the basis for understanding image features. To supply tokens for the Transformer block, a Conv-Transformer tokenizer (C-T tokenizer) is proposed to complete the dimension transformation between the feature map and token sequence, as well as extract local features. As shown in Fig. 3, given the input $\boldsymbol{F_s}^{i-1} \in \mathbb{R}^{H_{i-1}\times W_{i-1}\times C_{i-1}}$ of the C-T tokenizer in ith layer, we first employ the convolution layer to extract local features $\text{Conv2D}(\boldsymbol{F_s}^{i-1}) \in \mathbb{R}^{H_i\times W_i\times C_i}$. Then, each pixel in the feature map is encoded as a token, and the token sequence $\boldsymbol{T}_s^i = \{\boldsymbol{t}_1^s, \boldsymbol{t}_2^s, \ldots, \boldsymbol{t}_{H_i\cdot W_i}^s\}$ is obtained to feed into the Transformer block. In this way, the transformation of feature dimensions between the CNN and Transformer is available. The formula of this process is described as follows:

$$\boldsymbol{T}_s^i = \text{flatten}\big(\text{Conv2D}\big(\boldsymbol{F_s}^{i-1}, s\big)\big). \tag{18}$$

Different from the traditional token embedding of ViT, the developed C-T tokenizer can perceive the spatial information of local features. The local features from CNN are gradually fed back to the C-T tokenizer, which aims to enrich the local detail perception ability of the Transformer block. Besides, setting the output dimension and stride of convolution in the C-T tokenizer can regulate the dimension and number of tokens.

In the Transformer, the attention mechanism is utilized to model the context dependencies of the token sequence and capture the global information of features. To further strengthen the ability to learn local spatial features, we utilize CNN to replace the linear projection in the traditional attention mechanism structure. Specifically, in the multihead attention module, the token sequence of the ith layer is reshaped into a feature map, and then the convolution layer is used to obtain the spatial representation of tokens. After that, the feature map

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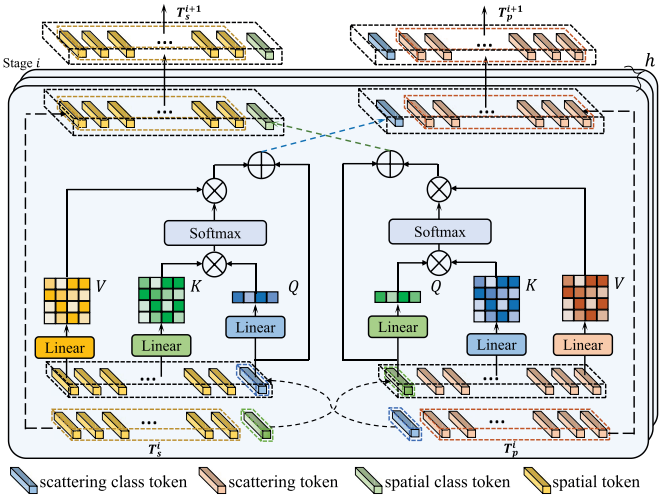


Fig. 4. Schematic architecture of SSIM. The left and right sides of this figure, respectively, describe the feature interaction process of SFEB and SAB branches in SSIM at stage i. Finally, the SSIM outputs the token sequences $\boldsymbol{T_s}^{i+1}$ and $\boldsymbol{T_p}^{i+1}$ of the two branches after information exchange.

is flattened back to the 1-D feature vector. Finally, the key matrix $\boldsymbol{K}$, query matrix $\boldsymbol{Q}$, and value matrix $\boldsymbol{V}$ are available. The formula expression of this process is as follows:

$$\boldsymbol{q}_j^i \backslash \boldsymbol{k}_j^i \backslash \boldsymbol{v}_j^i = \text{flatten}\Big(\text{Conv2D}_i^{q\backslash k\backslash v}\big(\text{reshape}\big(\boldsymbol{T}_s^i\big)\big)\Big) \tag{19}$$

where $\boldsymbol{T}_s^i$ represents the token sequence of the ith layer and $\boldsymbol{q}_j^i \backslash \boldsymbol{k}_j^i \backslash \boldsymbol{v}_j^i$ denotes the jth vector with respect to $\boldsymbol{Q}^i$, $\boldsymbol{K}^i$, and $\boldsymbol{V}^i$, respectively. The spatial attention mechanism of SFEB can be described as

$$\begin{aligned} \text{head}_h &= \text{Attention}\big(\boldsymbol{Q}_h^i, \boldsymbol{K}_h^i, \boldsymbol{V}_h^i\big) \\ &= \text{Softmax}\left(\frac{\boldsymbol{Q}_h^i \boldsymbol{K}_h^{i\,T}}{\sqrt{d_i}}\right)\boldsymbol{V}_h^i \end{aligned} \tag{20}$$

$$\text{MultiHead}_i = \text{Concat}(\text{head}_1, \text{head}_2, \ldots, \text{head}_h)\boldsymbol{W}_i^O \tag{21}$$

where head_h is the output of the hth head, d_i represents the normalization factor, and the value of d_i is equal to the feature dimension of the token. $\boldsymbol{W}_i^O$ represents the weight parameter of the full connection layer. In addition, we introduce the learned class token in spatial attention, which summarizes the semantic characteristics of all tokens.

In the proposed SFEB, the local detail features and the long-range context dependency of PolSAR images can be effectively extracted by this architecture composed of the CNN and Transformer.

C. Scattering-Spatial Interaction Module

After the feature extraction of two branches, scatter-aware features as well as hierarchical spatial features are obtained. To enhance the discrimination of these features from two branches, an SSIM is proposed to adaptively integrate these characteristics for cooperative classification. This module is arranged at the end of each hierarchical stage to exchange information between these two branches. The schematic architecture of SSIM is shown in Fig. 4.

Due to the significant semantic differences between the two branches, traditional feature concatenation or fusion methods are difficult to effectively fuse features. To overcome this issue, a cross-attention-based structure is adopted in the SSIM. Cross-attention structure can interact with semantic information between cross-branch tokens, which alleviates the imbalance of feature semantic information. Inspired by ViT, the class token aggregates the information of all patch tokens, so it is conducive to information exchange between branches and further enriches the feature expression of all tokens.

Specifically, the token set from SAB is denoted as $\boldsymbol{T}^p = \{\boldsymbol{t}_{\text{cls}}^p, \boldsymbol{t}_1^p, \boldsymbol{t}_2^p, \ldots, \boldsymbol{t}_{n_p}^p\}$, and the token set from SFEB is denoted as $\boldsymbol{T}^s = \{\boldsymbol{t}_{\text{cls}}^s, \boldsymbol{t}_1^s, \boldsymbol{t}_2^s, \ldots, \boldsymbol{t}_{n_s}^s\}$. We first input the class token of one branch and all the patch tokens of another branch into the cross-attention module to realize information interaction. Taking the information interaction of SAB as an example, the process can be described as

$$\boldsymbol{Q} = \boldsymbol{W}^q \boldsymbol{t}_{\text{cls}}^p, \quad \boldsymbol{K} = \boldsymbol{W}^k \boldsymbol{T}_{\text{in}}, \quad \boldsymbol{V} = \boldsymbol{W}^v \boldsymbol{T}_{\text{in}} \tag{22}$$

$$\begin{aligned} \text{CA}(\boldsymbol{T}_{\text{in}}) &= \text{CrossAttention}(\boldsymbol{Q}, \boldsymbol{K}, \boldsymbol{V}) \\ &= \text{Softmax}\left(\frac{\boldsymbol{Q}\boldsymbol{K}^T}{\sqrt{d_k}}\right)\boldsymbol{V} \end{aligned} \tag{23}$$

where $\boldsymbol{T}_{\text{in}} = \{\boldsymbol{t}_{\text{cls}}^p, \boldsymbol{t}_1^s, \boldsymbol{t}_2^s, \ldots, \boldsymbol{t}_{n_s}^s\}$ denotes the input tokens of the cross attention module, and $\boldsymbol{W}^q$, $\boldsymbol{W}^k$, $\boldsymbol{W}^q$ denote three learnable weight matrices, respectively. Then, the output $\boldsymbol{I}_{\text{cls}}^p \in \mathbb{R}^d$ after the interaction is obtained through residual connection, which can be described as follows:

$$\boldsymbol{I}_{\text{cls}}^p = \boldsymbol{t}_{\text{cls}}^p + \text{CA}(\boldsymbol{T}_{\text{in}}). \tag{24}$$

In the same way, the information interaction results $\boldsymbol{I}_{\text{cls}}^s$ of the SFEB is also available.

Since the feature representation form of the attention-based HSSIT is a token sequence, it provides feasibility for dynamically aggregating characteristics between two feature extraction branches. In the process of information exchange between two branches, the cross-attention mechanism plays an important role. This SSIM can employ scattering (or spatial) class tokens from one branch as query vector q to access all tokens from another branch. After cross attention mechanism in a contextual aware manner, the semantic information of class tokens from two branches will be supplemented, making the features more discriminative.

D. Hierarchical Network Architecture of HSSIT

As shown in Fig. 1, the proposed HSSIT is composed of multiple hierarchical stages, each of which consists of three components: an SAB, an SFEB, and an SSIM. The SAB is employed to capture the correlation between different polarimetric features. The SFEB is utilized to extract the hierarchical spatial features by iteratively using CNN and Transformer. The C-T tokenizer is responsible for regulating the feature dimensions and number of tokens at each stage. Through the feature extraction of SAB and SFEB, tokens will be input into SSIM for information exchange. After SSIM in each hierarchical stage, the semantic information of the class tokens from two branches can be supplemented. Then, we concatenate the scattering class token and spatial class token from HSSIT

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TABLE I

DETAILED NETWORK PARAMETER SETTINGS OF HSSIT. LINEAR PROJ: LINEAR PROJECTION LAYER, MHSA: MULTIHEADED SELF-ATTENTION, MHCA: MULTIHEADED CROSS-ATTENTION, LN: LAYER NORMALIZATION, DIM: THE DIMENSION OF FEATURE EMBEDDING, HEAD: THE NUMBER OF ATTENTION HEAD, MLP: MULTILAYER PERCEPTRON, N_{CLS}: NUMBER OF CATEGORIES

(Parameters)	Layer Name	Stage-1 (0.15M)	Stage-2 (0.76M)	Stage-3 (6.2M)
SAB	Linear Proj MHSA	[64-d, LN] (6.40K) [dim 64, head 2] ×1 (33.28K)	[128-d, LN] (8.32K) [dim 128, head 4] ×2 (0.21M)	[256-d, LN] (33.02K) [dim 256, head 8] ×4 (2.45M)
SFEB	C-T tokenizer Conv MHSA	[3×3, 64-d, stride 1] (15.74K) [3×3, 64-d dim 64, head 2] ×1 (35.78K)	[3×3, 128-d, stride 1] (74.11K) [3×3, 128-d dim 128, head 4] ×2 (0.27M)	[3×3, 256-d, stride 2] (0.29M) [3×3, 256-d dim 256, head 8] ×4 (2.68M)
SSIM	MHCA	[dim 64, head 2] ×1 (49.79K)	[dim 128, head 4] ×1 (0.19M)	[dim 256, head 8] ×1 (0.77M)
Classifier	MLP	[dim 512, 512, N_{cls}] (8.72K)		

for the joint classification. The mathematical expression of this process is as follows:

$$\hat{y} = \mathrm{MLP}\big(\mathrm{LN}\big(\mathrm{cat}\big(\boldsymbol{I}^{p}_{\mathrm{cls}}, \boldsymbol{I}^{s}_{\mathrm{cls}}\big)\big)\big) \tag{25}$$

where MLP(·) represents the multilayer perceptron model as a classifier and LN(·) represents layer normalization. Finally, cross-entropy loss is applied to optimize our model.

The parameter settings of the proposed HSSIT with three hierarchical stages are listed in Table I, which includes the number of attention blocks on three components, the number of attention heads, the projection dimensions of the linear layer and convolution layer, the number of hidden nodes in MLP, and the number of parameters for each stage. It can be seen from the table that as the stages deepen, the feature dimension gradually increases, which is conducive to learning richer semantic information. In stage 3, the C-T tokenizer (stride = 2) will reduce the size of the feature map, resulting in a decrease in the number of tokens. This can appropriately reduce the number of model parameters to improve the inference efficiency of the model.

IV. EXPERIMENTS

In this section, to analyze the superior performance of the proposed HSSIT, a series of experiments have been carried out on three PolSAR datasets. The effectiveness and feasibility of our model are analyzed and discussed.

A. Datasets

To evaluate the classification accuracies of the proposed model, three widely used PolSAR datasets are applied in our work, which are the Flevoland dataset, the San Francisco dataset, and the Oberpfaffenhofen dataset. The first two datasets are derived by the AIR-SAR system and belong to the sensor of L-band data. The last dataset is acquired by the ESAR airborne platform and also corresponds to the L-band. Figs. 5–7 shows their RGB image of Pauli decomposition and ground-truth image, respectively.

1) Flevoland Dataset: The ground resolution of the Flevoland Dataset is 6.6 × 12.1 m, and the size of this full-polarimetric image is 750 × 1024 pixels [45]. In this dataset, there are 15 categories, as shown in Fig. 5.

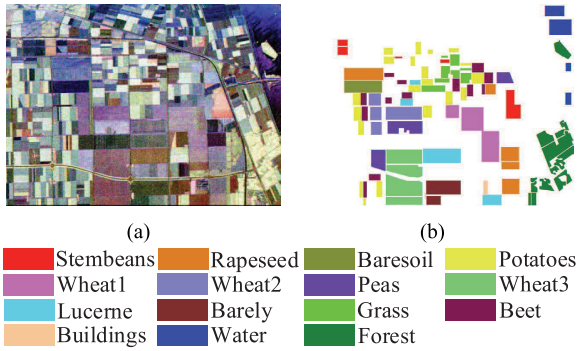


Fig. 5. Flevoland dataset. (a) Image obtained by Pauli decomposition. (b) Ground-truth image.



Fig. 6. San Francisco dataset. (a) Image obtained by Pauli decomposition. (b) Ground-truth image.

2) San Francisco Dataset: This dataset is also a polarimetric SAR image, whose spatial resolution is 10 m. This dataset contains 900 × 1024 pixels that are divided into four categories. As shown in Fig. 6, these categories include vegetation, water, low-density urban, and high-density urban [46], [47].

3) ESAR Oberpfaffenhofen Dataset: This is a full-polarimetric image, which is obtained from Oberpfaffenhofen, Germany. And the size of this image is 1100 × 1024 pixels. Each pixel of this image is classified into three kinds of ground objects. As shown in Fig. 7, these categories contain the built-up area, woodland, and open area [48].

B. Experimental Settings

All experiments are carried out on a high-performance server with an NVIDIA GeForce GTX 3090 GPU. All the algorithm implementation processes are completed on the python3.7 platform. The detail experimental settings of our

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(a) (b)

Fig. 7. ESAR Oberpfaffenhofen dataset. (a) Image obtained by Pauli decomposition. (b) Ground-truth image.

TABLE II
POLARIMETRIC INPUT FEATURES USED IN THE EXPERIMENTS

Decomposition	Feature details	Dimension
Pauli	odd, even, $\pi/4$ rotated even scattering	3
Krogager	sphere, diplane, helix	3
$H/A/\alpha$	alpha, anisotropy, beta, delta, entropy, gamma, lambda combination: HA, (1-H)A, H(1-A), (1-H)(1-A)	11
Yamaguchi	Yamaguchi4:(Odd,Dbl,Vol, Hix)dB	4

proposed method as well as comparative methods will be introduced in the following.

In our experiments, Pauli, Krogager, $H/A/\alpha$, and Yamaguchi polarimetric decompositions are applied to obtain polarimetric features, which are listed in Table II. In our proposed HSSIT, the number of hierarchical stages is set to 3, and each stage is used to extract different levels of features. The feature dimensions of the three hierarchical stages are set to 64, 128, and 256, respectively. And the feature dimension of each attention head is set to 32. More detail parameter settings of each architecture stage are listed in Table I. Stochastic gradient descent (SGD) optimizer is utilized for training the network, where the learning rate is set to 0.001 for 250 epochs, the weight decay is set to 0.0005, and the momentum is set to 0.9.

For the experiments on the Flevoland dataset, we randomly select 2% labeled pixels per land cover category as the training samples. Besides, 1% and 97% of the sample pixels are randomly chosen to use for the validation set and test set, respectively. As for the San Francisco dataset and Oberpfaffenhofen dataset, 5%, 1%, and 94% of the sample pixels from each land cover category in the image are used for model training, validation, and testing, respectively. In detail, we take a sample pixel as the center and then crop an image patch with a certain size around it as the input of the model. The land cover category of the center pixel is annotated as the label of the image patch.

In our work, several competitive methods for PolSAR image classification have been used for comparative experiments, which aim to demonstrate the effectiveness of our proposed method. According to the structure of the network, all methods in the experiment can be divided into two types. The first one belongs to the CNN-based model, which contains 2DCNN [31], 3DCNN, and attention-based polarimetric feature selection convolutional network (AFS-CNN) [20]. The other type is the Transformer-based method, including the ViT [35] and pyramid ViT (PVT) [49]. In the following, the specific experimental settings for these comparative algorithms are described.

1) 2DCNN: It consists of two convolutional layers, a BN layer, a rectified linear unit, a max pooling layer, and a fully connected layer. On this basis, we added an extra convolutional layer to project the high-dimensional input features. To obtain the optimal classification performance, the patch size for the input of 2DCNN is set to 9×9.

2) 3DCNN: The structure of 3DCNN is similar to that of 2DCNN, except for extending the convolutional kernel to 3-D. The other parameter settings remain consistent with 2DCNN.

3) Attention-Based Polarimetric Feature Selection Convolutional Network: To ensure the effectiveness of high-dimensional data classification, AFS-CNN introduces an attention-based structure to capture the relationship between polarimetric features. After the attention-based module, six convolutional layers are employed to continue extracting features. To achieve the best classification accuracies, the input patch size is set to 15×15.

4) ViT: It is constructed by multiple Transformer blocks, and the depth of the network is set to 6. For the settings of multihead self-attention, the number of heads is 4 and the feature embedding dimension of each head is set to 32. To yield superior results, the number of hidden nodes in MLP is set to 256, and the patch size is set to 14×14.

5) Pyramid Vision Transformer: Similar to our proposed HSSIT, PVT also belongs to a hierarchical structure with multiple Transformer blocks. At the beginning of each stage, use a convolutional or pooling layer for downsampling to reduce the number of tokens. The feature embedding dimensions of the three hierarchical stages are set to [64, 128, 320]. The depth of the Transformer for each stage is set to [3, 6, 3], and the number of attention heads is set to [2, 4, 8]. The patch size for the input is set to 20×20.

In comparative experiments, four quantitative metrics are adopted to evaluate the classification performance of all methods, which include overall accuracy (OA), average accuracy (AA), kappa coefficient, and per-class accuracy. Moreover, classification maps are also used to evaluate classification performance from a visual perspective.

C. Experimental Comparison With Competitive Methods

Classification results of compared models and proposed model on the three datasets are reported in Tables III–V. These tables report the specific results of the comparative experiments, including OA (%), AA (%), and Kappa (ranging from 0 to 1) of each model, as well as the accuracy of each category (%). The values marked in bold indicate the maximum value for this row. The complexity, training time, and inference time of each model are also listed in these tables. Figs. 8–10 show classification maps of each model on three datasets, which are used to visually display the classification performance of all models. The compared results on each dataset are analyzed as follows.

1) Results on the Flevoland Dataset: As seen from the classification maps in Fig. 8, our proposed model has better classification performance than the compared models. The classification maps of CNN-based methods are visually shown

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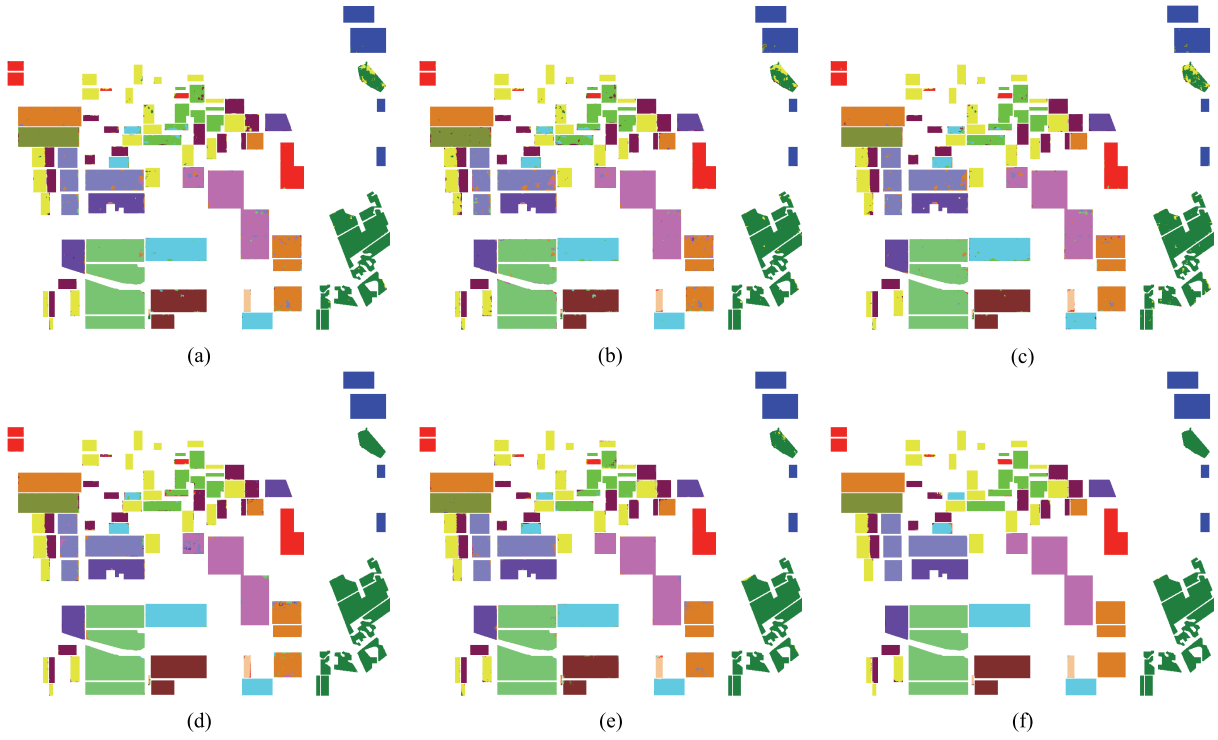


Fig. 8. Visualization of classification maps on the Flevoland dataset. (a) 2DCNN, (b) 3DCNN, (c) AFS-CNN, (d) ViT, (e) PVT, and (f) HSSIT.

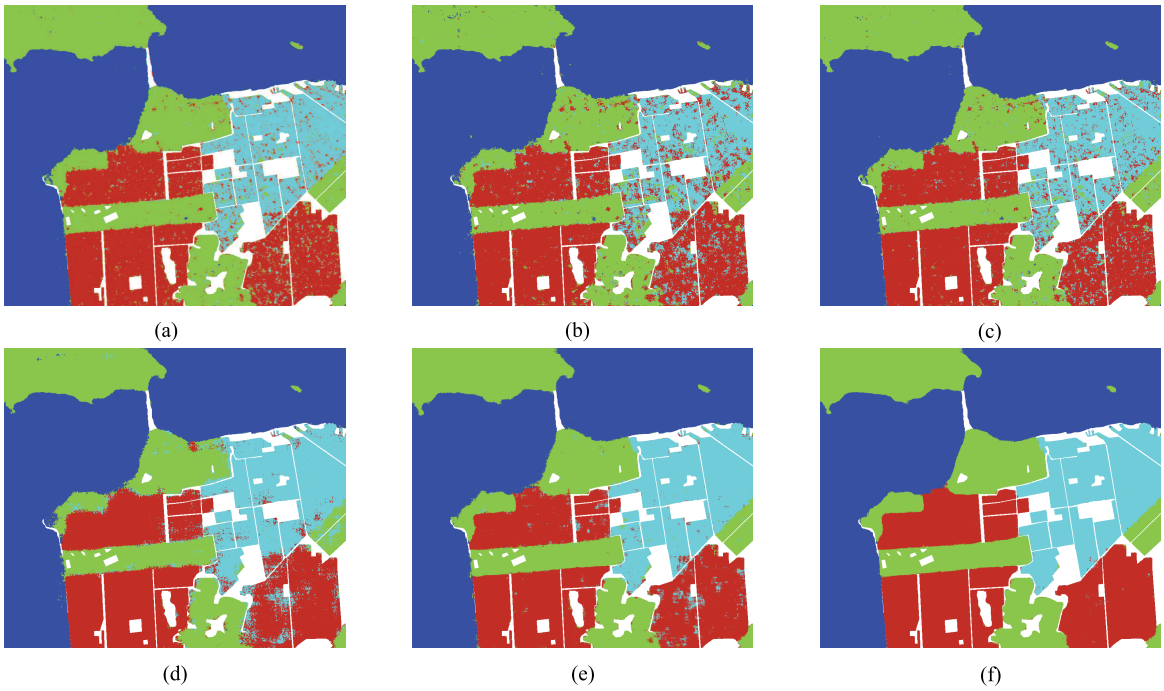


Fig. 9. Visualization of classification maps on the San Francisco dataset. (a) 2DCNN, (b) 3DCNN, (c) AFS-CNN, (d) ViT, (e) PVT, and (f) HSSIT.

in Fig. 8(a)–(c), and they produced more discrete misclassified pixels, which may be caused by the limited receptive field of the CNN. Especially in areas of wheat2, a large number of pixels are incorrectly classified to the rapeseed category. On the contrary, the Transformer-based methods perform better. Since the Transformer can effectively capture context dependencies among pixels from the PolSAR image patch, confusion over large areas between different categories is not common. Compared with other methods, the proposed HSSIT can spatially aggregate features of pixels from the same category, and the scatter-aware features supplied by the SAB can enable the model to obtain more discriminative features, which can promote classification performance. From the results of visualization, the proposed method can yield fewer misclassified pixels on the boundaries than ViT and PVT.

As reported in Table III, the proposed method achieves the best performance on all three overall criteria compared with other methods. There are 1.12%, 1.66%, and 0.0122%

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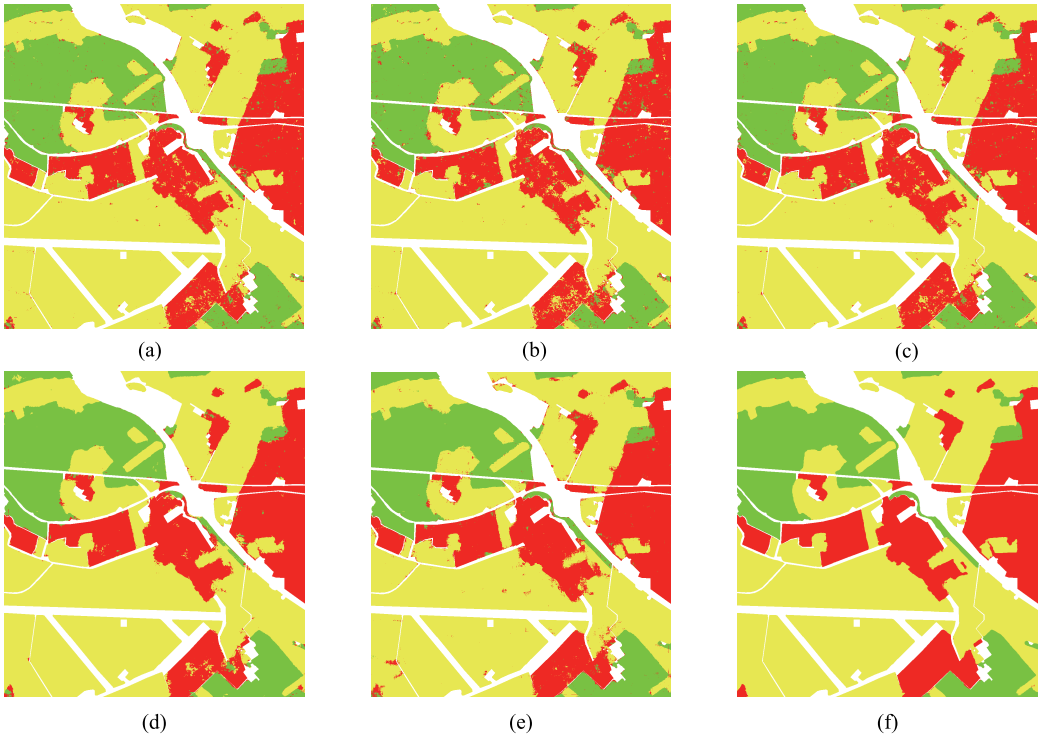


Fig. 10. Visualization of classification maps on the ESAR Oberpfaffenhofen dataset. (a) 2DCNN, (b) 3DCNN, (c) AFS-CNN, (d) ViT, (e) PVT, and (f) HSSIT.

increases of OA, AA, and Kappa coefficient compared with the best of comparing methods, respectively. Moreover, for the categories of wheat2, beets, and buildings, accuracies obtained by the proposed method are significantly higher than other classification models.

2) Results on San Francisco Dataset: Classification maps of all classification models on the San Francisco dataset are shown in Fig. 9. It is observed that the classification map of the proposed HSSIT is closer to the ground-truth map. Compared to the Flevoland dataset, this dataset has fewer land cover categories, but more complex scattering characteristics of each category. It may be the reason why other compared methods have poor performance for the categories of low-density urban and high-density urban. Benefiting from the SAB and SSIM, our proposed HSSIT can handle more complex polarimetric features to yield a smoother classification map than the compared methods.

As seen in Table IV, our method yields the highest evaluation criteria of OA, AA, and Kappa coefficient. There are 1.39%, 2.18%, and 0.0198% increases for OA, AA, and Kappa coefficient compared with the proposed HSSIT and the best of comparing methods, respectively. In addition, the proposed HSSIT can effectively distinguish low-density urban areas and high-density urban areas, which is not well done by other methods. Especially for the classification of high-density urban areas, the OAs obtained by CNN-based methods are lower than 90%. This may mean that the global information extracted by the Transformer is beneficial for SAR image classification.

3) Results on the Oberpfaffenhofen Dataset: Fig. 10 shows the classification maps obtained by several comparative methods and the proposed HSSIT on Oberpfaffenhofen dataset. It can be observed that the classification map of our proposed HSSIT is smooth, with almost no misclassification, while CNN-based methods have significant misclassifications. Since the areas of different land cover categories are located adjacent to each other in spatial positions, the compared methods misclassify a large number of pixels belonging to built-up areas (red) into open areas (yellow). Compared to other methods, our method performs better when dealing with pixels located at the boundaries of adjacent categories.

Experimental results of all classification methods are reported in Table V. As far as OA, AA, and kappa are concerned, the proposed model achieves the best results and improves three criteria by 0.61%, 0.62%, and 0.0102% compared with the best of comparing methods. In addition, the proposed HSSIT achieves the highest accuracy in each category, 99.14% in built-up areas, 99.39% in open areas, and 99.14% in wood land, respectively.

From the above experimental results, it can be observed that deeper models have a larger number of model parameters. Transformer-based models have more complex structures as well as more parameters compared to CNN-based models. Our proposed HSSIT flexibly combines the structure of the CNN and Transformer, which reduces a certain amount of parameters. However, the dual branch structure and cross-attention structure result in more model parameters than PVT. In addition, lightweight CNN-based models have faster inference speed and shorter training time. Due to the largest number of training samples in the ESAR Oberpfaffenhofen dataset, the training time of all models has significantly increased. From the classification accuracy indicators of the model on three PolSAR datasets, it can be seen that the Transformer-based models achieve better classification performance than the CNN-based methods, and the classification performance of our proposed HSSIT is superior to other Transformer-based methods. It can be observed that more complex models

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TABLE III
EXPERIMENTAL RESULTS OF ALL CLASSIFICATION METHODS ON THE FLEVOLAND DATASET

Class	2DCNN	3DCNN	AFS-CNN	ViT	PVT	HSSIT
Steam beans	97.93 ± 0.04	97.71 ± 0.35	97.86 ± 0.12	97.40 ± 0.12	98.58 ± 0.12	**99.65 ± 0.08**
Rapeseed	95.30 ± 0.11	91.44 ± 0.28	96.57 ± 0.14	95.11 ± 0.13	96.63 ± 0.23	**98.44 ± 0.09**
Bare soil	97.69 ± 0.03	91.78 ± 0.31	99.00 ± 0.12	96.99 ± 0.14	99.10 ± 0.09	**99.59 ± 0.11**
Potatoes	98.28 ± 0.09	97.14 ± 0.25	98.29 ± 0.18	**99.29 ± 0.09**	98.29 ± 0.18	98.78 ± 0.22
Wheat1	95.60 ± 0.17	95.54 ± 0.30	93.94 ± 0.17	95.71 ± 0.18	98.80 ± 0.12	**99.89 ± 0.07**
Wheat2	94.29 ± 0.18	88.12 ± 0.32	92.33 ± 0.11	94.38 ± 0.12	97.23 ± 0.10	**99.87 ± 0.08**
Peas	96.99 ± 0.13	96.71 ± 0.20	97.27 ± 0.08	98.66 ± 0.12	98.74 ± 0.21	**99.23 ± 0.12**
Wheat3	99.15 ± 0.04	97.34 ± 0.29	98.71 ± 0.04	99.52 ± 0.07	99.43 ± 0.14	**99.95 ± 0.03**
Lucerne	97.54 ± 0.19	94.99 ± 0.28	94.20 ± 0.12	98.38 ± 0.11	97.42 ± 0.18	**99.14 ± 0.14**
Barley	97.69 ± 0.15	95.60 ± 0.31	98.86 ± 0.09	99.75 ± 0.02	98.72 ± 0.17	**99.80 ± 0.13**
Grasses	92.50 ± 0.12	90.01 ± 0.35	96.60 ± 0.11	96.52 ± 0.17	95.91 ± 0.22	**99.06 ± 0.11**
Beets	95.88 ± 0.15	95.96 ± 0.21	98.02 ± 0.04	96.61 ± 0.12	97.81 ± 0.26	**98.61 ± 0.11**
Buildings	91.14 ± 0.15	87.04 ± 0.37	86.16 ± 0.12	87.82 ± 0.12	91.14 ± 0.24	**99.92 ± 0.05**
Water	99.79 ± 0.05	97.70 ± 0.12	98.98 ± 0.11	99.87 ± 0.09	99.00 ± 0.07	**99.91 ± 0.07**
Forest	96.75 ± 0.13	93.76 ± 0.23	94.50 ± 0.12	**99.85 ± 0.10**	99.03 ± 0.18	99.73 ± 0.12
OA (%)	96.88 ± 0.09	94.70 ± 0.29	96.55 ± 0.11	97.81 ± 0.13	98.29 ± 0.16	**99.41 ± 0.11**
AA (%)	96.43 ± 0.11	94.06 ± 0.31	96.09 ± 0.13	97.07 ± 0.11	97.79 ± 0.18	**99.45 ± 0.13**
Kappa	0.9660 ± 0.0010	0.9422 ± 0.0033	0.9624 ± 0.0015	0.9761 ± 0.0014	0.9813 ± 0.0015	**0.9935 ± 0.0014**
Parameters (M)	1.14	5.46	6.60	7.93	6.68	7.21
Inference time (ms)	0.61	0.68	2.10	5.95	9.37	9.05
Training time (s)	32.09	50.96	62.57	793.58	572.89	661.48

TABLE IV
EXPERIMENTAL RESULTS OF ALL CLASSIFICATION METHODS ON THE SAN FRANCISCO DATASET

Class	2DCNN	3DCNN	AFS-CNN	ViT	PVT	HSSIT
Water	99.60 ± 0.06	99.36 ± 0.24	99.54 ± 0.14	99.47 ± 0.08	**99.76 ± 0.06**	99.46 ± 0.17
Vegetation	94.51 ± 0.16	92.58 ± 0.39	94.36 ± 0.21	95.36 ± 0.06	97.82 ± 0.13	**99.33 ± 0.18**
Low-density urban	94.48 ± 0.12	89.25 ± 0.32	92.90 ± 0.22	93.34 ± 0.15	97.67 ± 0.19	**99.21 ± 0.16**
High-density urban	71.59 ± 0.27	70.46 ± 0.42	82.44 ± 0.31	92.36 ± 0.14	93.76 ± 0.18	**99.74 ± 0.11**
OA(%)	95.84 ± 0.17	91.56 ± 0.39	94.51 ± 0.21	96.15 ± 0.12	98.02 ± 0.17	**99.41 ± 0.10**
AA(%)	94.43 ± 0.19	87.91 ± 0.32	92.31 ± 0.23	95.13 ± 0.14	97.25 ± 0.18	**99.43 ± 0.13**
Kappa	0.9415 ± 0.0020	0.8812 ± 0.0037	0.9228 ± 0.0029	0.9460 ± 0.0011	0.9722 ± 0.0015	**0.9917 ± 0.0012**
Parameters (M)	1.14	5.46	6.60	7.93	6.68	7.21
Inference time (ms)	0.62	0.71	2.11	5.98	9.41	9.11
Training time (s)	629.55	894.74	914.56	3056.88	2132.12	2792.41

TABLE V
EXPERIMENTAL RESULTS OF ALL CLASSIFICATION METHODS ON THE ESAR OBERPFAFFENHOFEN DATASET

Class	2DCNN	3DCNN	AFS-CNN	ViT	PVT	HSSIT
Bulit-up areas	94.87 ± 0.22	90.82 ± 0.31	94.13 ± 0.28	95.91 ± 0.16	97.99 ± 0.29	**99.14 ± 0.11**
Open areas	98.08 ± 0.12	97.84 ± 0.41	98.12 ± 0.26	98.56 ± 0.21	98.87 ± 0.22	**99.39 ± 0.09**
Wood land	96.88 ± 0.27	95.70 ± 0.40	97.08 ± 0.33	95.81 ± 0.20	98.95 ± 0.24	**99.14 ± 0.14**
OA(%)	97.08 ± 0.21	95.74 ± 0.35	96.96 ± 0.30	97.37 ± 0.18	98.67 ± 0.20	**99.28 ± 0.13**
AA(%)	96.61 ± 0.23	94.78 ± 0.32	96.44 ± 0.29	96.76 ± 0.19	98.60 ± 0.21	**99.22 ± 0.12**
Kappa	0.9505 ± 0.0021	0.9276 ± 0.0033	0.9485 ± 0.0031	0.9553 ± 0.0018	0.9776 ± 0.0019	**0.9878 ± 0.0013**
Parameters (M)	2.21	6.26	6.60	7.93	6.68	7.21
Inference time (ms)	0.74	0.83	2.23	6.01	9.53	9.19
Training time (s)	992.47	1487.88	1759.87	5869.59	4092.41	4896.94

typically yield higher classification accuracy. In summary, our proposed HSSIT has fewer parameters compared to ViT and achieves the highest classification accuracy on three PolSAR datasets.

D. Discussions

To further analyze the performance of the proposed model, experiments including ablation studies, hyperparameter analyses, and sampling strategies are introduced in this section.

1) Ablation Studies: Our proposed HSSIT is composed of SAB, spatial features extraction branch (SFEB), and SSIM.

TABLE VI
CLASSIFICATION ACCURACIES WITH DIFFERENT COMBINATIONS ON THREE DATASETS

SAB	SFEB	SSIM	Flevoland	San Francisco	Oberpfaffenhofen
√			96.06	96.43	96.05
	√		99.31	99.19	99.09
√	√		99.34	99.25	99.17
√	√	√	**99.41**	**99.41**	**99.28**

Since the excellent classification performance of HSSIT benefits from these modules, we conduct ablation studies to analyze the contributions of each module to PolSAR image

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TABLE VII
EXPERIMENTAL RESULTS OF ALL CLASSIFICATION METHODS ON THE FLEVOLAND DATASET UNDER CLUSTER SAMPLING

Class	2DCNN	3DCNN	AFS-CNN	ViT	PVT	HSSIT
Steam beans	89.41 ± 1.32	84.71 ± 1.35	**91.95 ± 0.92**	89.21 ± 1.12	88.83 ± 1.12	87.15 ± 1.38
Rapeseed	97.85 ± 0.41	95.63 ± 0.85	93.73 ± 0.74	98.17 ± 0.93	94.90 ± 0.83	**98.31 ± 0.79**
Bare soil	97.71 ± 0.73	95.40 ± 1.31	95.97 ± 0.62	99.40 ± 0.74	**99.78 ± 0.09**	98.44 ± 0.71
Potatoes	98.50 ± 0.33	96.12 ± 0.75	96.14 ± 0.78	98.63 ± 0.49	98.14 ± 0.68	**98.70 ± 0.32**
Wheat1	84.81 ± 1.17	83.00 ± 2.02	82.29 ± 1.67	86.15 ± 1.18	85.11 ± 2.01	**86.52 ± 1.47**
Wheat2	61.64 ± 2.18	64.72 ± 2.32	66.88 ± 2.11	**78.73 ± 2.12**	75.90 ± 2.10	72.11 ± 2.08
Peas	99.32 ± 0.13	97.91 ± 1.23	99.13 ± 0.38	99.48 ± 0.22	99.50 ± 0.21	**99.59 ± 0.12**
Wheat3	61.96 ± 2.84	65.68 ± 2.29	71.11 ± 1.54	62.00 ± 2.07	75.32 ± 2.14	**82.90 ± 1.93**
Lucerne	67.17 ± 2.19	66.28 ± 2.28	67.71 ± 1.72	**68.84 ± 2.11**	63.60 ± 2.18	65.25 ± 2.14
Barley	98.57 ± 0.15	95.48 ± 1.31	96.30 ± 0.69	98.92 ± 0.82	99.39 ± 0.17	**99.89 ± 0.13**
Grasses	94.69 ± 0.12	84.34 ± 1.35	86.29 ± 1.11	**98.62 ± 0.17**	91.12 ± 0.72	96.91 ± 0.91
Beets	94.67 ± 0.15	93.73 ± 1.21	89.94 ± 1.04	94.56 ± 0.72	**94.82 ± 0.26**	93.99 ± 0.61
Buildings	98.93 ± 0.15	99.36 ± 0.67	98.71 ± 0.72	98.50 ± 0.62	97.42 ± 0.74	**99.92 ± 0.05**
Water	98.03 ± 0.05	99.60 ± 0.22	87.85 ± 0.71	99.26 ± 0.19	99.10 ± 0.07	**99.75 ± 0.07**
Forest	75.79 ± 2.13	70.04 ± 1.23	72.82 ± 1.52	**78.52 ± 1.61**	69.40 ± 2.78	74.37 ± 2.12
OA (%)	85.28 ± 1.29	84.90 ± 1.44	85.05 ± 1.21	87.32 ± 1.14	88.50 ± 1.26	**90.40 ± 1.18**
AA (%)	87.86 ± 1.31	86.46 ± 1.31	86.69 ± 1.23	89.84 ± 1.12	88.87 ± 1.17	**90.26 ± 1.21**
Kappa	0.8381 ± 0.0120	0.8338 ± 0.0133	0.8353 ± 0.0125	0.8606 ± 0.0114	0.8733 ± 0.0120	**0.8941 ± 0.0119**

classification. Experimental results of ablation studies on three datasets are reported in Table VI. It can be observed that the accuracy of only using SAB is not very high. While, the overall accuracies of SFEB on all three datasets exceed 99.0%, which indicates that our proposed module can extract spatial features effectively. According to the performance of using the single module, spatial features make a great contribution to enhancing the classification accuracy. To analyze the performance of the combination of SAB and SFEB, we directly concatenate features of the two modules in the last stage for the classifier. When combining SAB and SFEB together, it can be seen that the accuracies improve slightly, owing to more semantic information obtained from spatial features and scatter features. When SSIM is added to a hierarchical structure, the OA on the three datasets increases by 0.07%, 0.16%, and 0.11%, respectively, which indicates that SSIM can effectively aggregate spatial tokens and scatter tokens to learn more discriminative features. Compared with using SFEB alone, the addition of SAB and SSIM results in a 0.10%, 0.22%, and 0.19% increase in model accuracy on the three datasets, respectively. These three modules work together to fully utilize spatial semantic information and polarimetric characteristics, achieving high classification accuracy.

2) Hyperparameter Analyses: Our proposed HSSIT model consists of multiple hierarchical structures with a Transformer, in which the depth of the model is related to feature extraction capabilities. The models with too deep a network structure often have a large amount of parameters, which can easily lead to over fitting during training. Too shallow a network may contribute to under fitting, and the feature representation capability will be affected. In addition, the number of heads in the multiheaded self-attention also affects the performance of the network. Since each attention head corresponds to a separate subspace, the number of attention heads is closely related to the information acquisition of the multiheaded self-attention mechanism. Therefore, the mixed experiments are conducted on the Flevoland dataset to obtain optimal settings, and the results are shown in Fig. 11. It is observed that appropriately increasing the depth and attention head of the network is beneficial for improving the ability of feature representation. However, excessive increases may cause information redundancy to impair classification performance. According to the experimental results, when the depth of the model is set to 10 (including seven blocks of SAB/SFEB parallel branch and three SSIM blocks) and the number of attention heads is set to 8, classification accuracy can achieve the best. To balance the semantic differences between SAB and SFEB, these two branches follow the same structural settings.

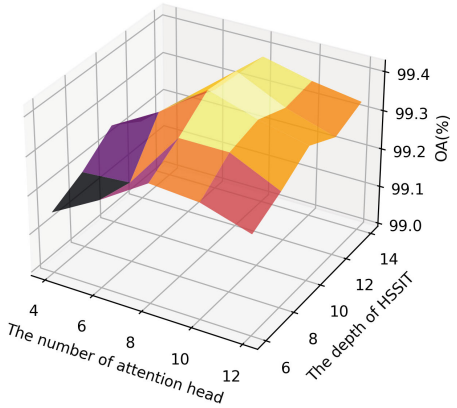


Fig. 11. Influence of different structural settings on classification performance. The number of attention heads corresponds to the setting of the third stage. In each stage, the number of SSIM blocks is fixed to 1, and the number of SAB and SFEB is equal. The candidates for the depth of SAB / SFEB in the three stages [d1, d2, d3] are set to [1, 1, 1], [1, 2, 2], [1, 2, 4], [1, 2, 6], [1, 4, 6], respectively. After adding SSIM, the candidate depth of HSSIT is 6, 8, 10, 12, and 14, respectively.

In our method, SAB and SFEB play different roles in classification, so the suitable sizes of input patches for these two branches need to be considered separately. Classification accuracies with different combinations of the patch size k for SAB and the patch size l for SFEB are reported in Fig.12, where k is ranging from 6 to 14, and l is ranging from 18 to 26. As shown in Fig. 12(a)–(c), the optimal combinations $[k, l]$ of the patch size for the flevoland dataset, San Francisco dataset, and Oberpfaffenhofen dataset are [10, 22], [10, 20], [8, 22], respectively.

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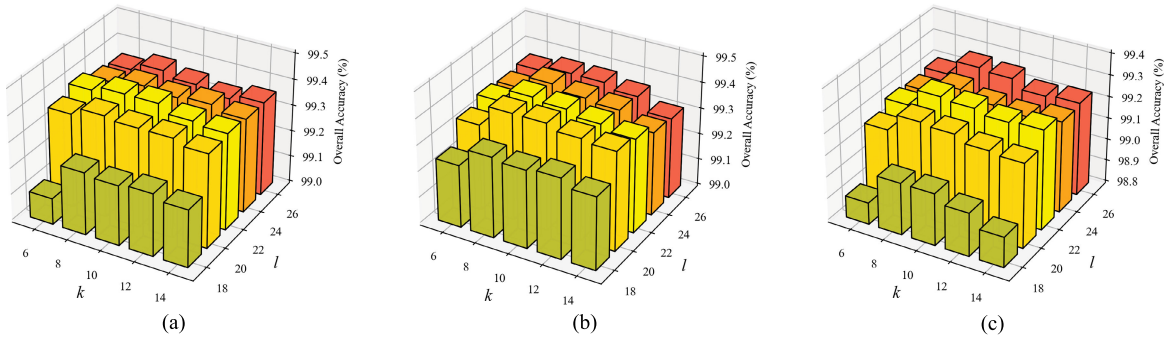


Fig. 12. OA of the HSSIT with different input size combinations [k, l] of SAB and SFEB on three datasets. k represents the input size for SAB, and l represents the input size for SFEB. (a) Flevoland dataset. (b) San Francisco dataset. (c) Oberpfaffenhofen dataset.

3) Analyses of Sampling Strategy: Although random sampling is the most popular sampling strategy for evaluating model performance in PolSAR image classification, it may lead to biased error when only one PolSAR image is available. For random sampling, two adjacent pixels can be selected as a test sample and a train sample. There is a large overlap between the image patches surrounding these pixels, which results in a spatial correlation between the test data and the training data [50]. To weaken this correlation, Hänsch et al. [51] proposed a cluster sampling to make train and test samples of the same class away from each other. This sampling strategy has been applied in our work to further evaluate the performance of all classification models. First, for each class, the spatial coordinates of the samples in the PolSAR image are clustered into two clusters. Then, we randomly select one cluster as the train data and the other cluster as the test data.

Experimental results of all classification methods on the Flevoland dataset under cluster sampling are reported in Table VII. It can be seen that our proposed model still yields the highest evaluation criteria of OA, AA, and Kappa coefficient. Besides, our proposed method achieves the highest classification accuracy on multiple categories. Overall, compared with other classification models, the proposed HSSIT obtains the best classification performance under both random sampling and cluster sampling cases.

V. Conclusion

In this article, an HSSIT for PolSAR image classification is proposed to make full use of deep spatial semantic information and polarimetric characteristics. In the multistage hierarchical structure of HSSIT, SAB aims to model the correlation between polarimetric features, and the spatial feature extracting branch can improve the global information perception ability for spatial features. After that, we further propose an information exchange module based on the cross attention to aggregate tokens from two branches, allowing the model to adaptively enhance feature representations. Comparative experiments on three PolSAR datasets demonstrate that the proposed HSSIT can yield better classification performance than CNN-based methods and other Transformer-based methods, and ablation studies certify the effectiveness of the proposed modules.

The proposed model applies the supervised learning method for PolSAR image classification. In future work, we will explore the self-supervised learning method to obtain a pretraining model with better generalization, making it possible to transfer the model to other classification tasks of PolSAR images.

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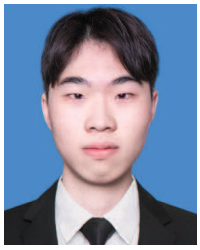
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Jie Geng (Member, IEEE) received the B.S. degree in electronic information engineering and the Ph.D. degree in signal and information processing from the Dalian University of Technology, Dalian, China, in 2013 and 2018, respectively.

He is currently an Associate Professor with the School of Electronics and Information, Northwestern Polytechnical University, Xi'an, China. His research interests include synthetic aperture radar (SAR) image processing, remote-sensing image classification, and few-shot learning.



Yuhang Zhang received the B.S. degree in communication engineering from the Wuhan Institute of Technology, Wuhan, China, in 2021. He is currently pursuing the M.S. degree in electronic information with the School of Electronics and Information, Northwestern Polytechnical University, Xi'an, China.

His research interests include PolSAR image classification.



Wen Jiang received the bachelor's and master's degrees from the Information Engineering University, Zhengzhou, China, in 1994 and 1997, respectively, and the Ph.D. degree from Northwestern Polytechnical University, Xi'an, China, in 2009.

She is currently a Professor with the School of Electronics and Information, Northwestern Polytechnical University. Her research interests include information fusion, artificial intelligence, remote-sensing image processing, and intelligent algorithm security.

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